



Hashing

1. Concept of Hashing
2. An important concept in hashing
3. Types of Hashing
4. Example
5. Collisions in hashing

Hashing is searching technique. **Hashing in the data structure** is a technique of mapping a large chunk of data into small tables using a hashing function. It is also known as the message digest function. It is a technique that uniquely identifies a specific item from a collection of similar items.

There are other searching techniques like linear search, binary search, search trees etc. In these techniques, time taken to search any particular element depends on the total number of elements.

Example-

Linear Search takes $O(n)$ time to perform the search in unsorted arrays consisting of n elements.

Binary Search takes $O(\log n)$ time to perform the search in sorted arrays consisting of n elements.

It takes $O(\log n)$ time to perform the search in **Binary Search Tree** consisting of n elements.

The main drawback of these techniques is-

As the number of elements increases, time taken to perform the search also increases. This becomes problematic when total number of elements becomes too large.

Hashing-

Hashing is a well-known technique to search for any particular element among several elements.

It minimizes the number of comparisons while performing the search.

Advantage-

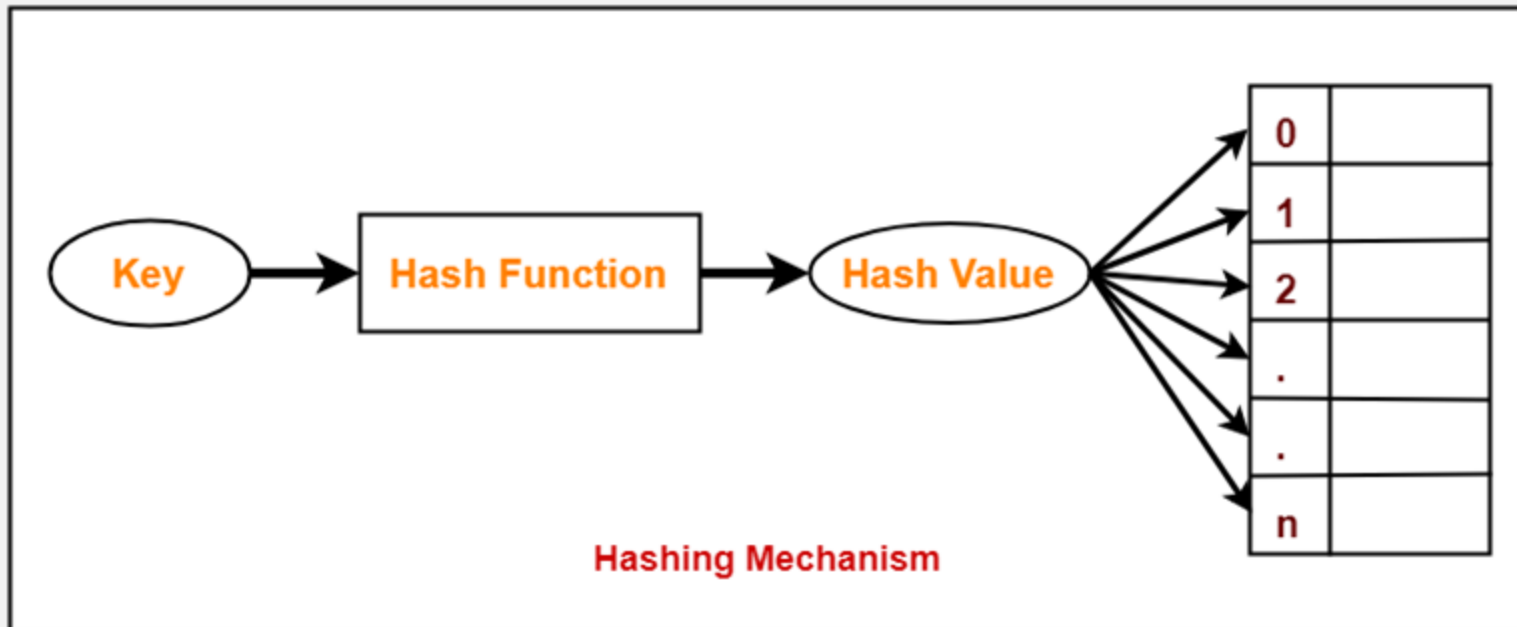
Unlike other searching techniques, Hashing is extremely efficient.

The time taken by it to perform the search does not depend upon the total number of elements.

It completes the search with constant time complexity $O(1)$.

Hashing Mechanism-

An array data structure called as **Hash table** is used to store the data items. Based on the hash key value, data items are inserted into the hash table.



Hash Key Value-

Hash key value is a special value that serves as an index for a data item. It indicates where the data item should be stored in the hash table. Hash key value is generated using a hash function.

Hash Function-

Hash function takes the data item as an input and returns a small integer value as an output.

The small integer value is called hash value.

Hash value of the data item is then used as an index for storing it into the hash table.

The properties of a good hash function are-

- It is efficiently computable.
- It minimizes the number of collisions.
- It distributes the keys uniformly over the table.

When the hash value of a key maps to an already occupied bucket of the hash table, it is called as a **Collision**.

Collision in Hashing-

In hashing,

- Hash function is used to compute the hash value for a key.
- Hash value is then used as an index to store the key in the hash table.
- Hash function may return the same hash value for two or more keys.

